PASHU SATHI — Government Command Dashboard Q&A

SIH 2026 5-Year Historical Judge Question Bank & Technical Defense (SIH26128)

Target System: PASHU SATHI Government Command Center (React 18 + PostGIS GIS)	**Deployment:** Live Vercel Production + Azure REST API
Primary Users: State Epidemiologists, District Magistrates & Veterinary Officers	**Evaluation Scope:** UI/UX, Spatial Engine, Actionability, Security & GIGW

Part 1: Front-End Architecture, GIS Engine & Performance (Technical)

Q1: Explain the dashboard tech stack. Why choose React + Vite over Angular or Next.js for a government tool?

Role: Frontend & GIS Engineer

• **Modern Component Core:** Built with React 18, TypeScript, and Vite for instant HMR development and minimal production bundle size (< 240 KB gzipped).

• **Tailwind CSS & Tactical Tokens:** Uses a strict, accessible design token system (Navy, Forest Green, 4-tier risk tokens) tailored for high readability in harsh daylight or emergency control rooms.

• **Vite Static Build Advantage:** Unlike heavy SSR frameworks like Next.js that require node servers, Vite outputs purely static optimized assets deployable to edge CDNs (Vercel / S3) with zero cold-start latency.

• **Testing Rigor:** Backed by 95 unit and integration tests using Vitest and React Testing Library (100% green).

Q2: How is the spatial map implemented? Why Leaflet instead of heavy Google Maps API?

Role: GIS & Mapping Specialist

• **Open-Source PostGIS Compliance:** Built with Leaflet.js natively consuming RFC 7946 GeoJSON layers directly emitted by PostgreSQL 17 / PostGIS 3.5.

• **Zero Commercial API Tax:** Google Maps incurs high per-load API costs and strict quota cutoffs. Leaflet pairs with CARTO Voyager / MapTiler vector tiles at near-zero operating cost.

• **Multi-Layer Vector Overlay:** Renders 394 administrative boundaries (1 State, 36 Districts, 357 Talukas), dynamic 25–50 km buffer perimeters, and kernel density estimation (KDE) heatmaps.

• **Dynamic Camera Auto-Fly:** Selecting any administrative scope (e.g. Pune, Khandesh, Baramati) executes programmatic viewport bounding-box fly animations.

Q3: How do you render 357 taluka boundary polygons without lagging on low-spec government PCs?

Role: Performance & Optimization Engineer

• **Zoom-Dependent Layer Toggling:** Taluka boundary GeoJSON is conditionally mounted only when current zoom level reaches >= 9 (`TALUKA_AUTO_ZOOM_THRESHOLD`), preventing DOM bloat at statewide view.

• **TopoJSON / Simplified Coordinate Precision:** Administrative boundary geometry coordinates are pre-simplified to 4 decimal places, reducing polygon vertex weight by over 65%.

• **R-Tree Spatial Caching:** Layers are managed in dedicated Leaflet `LayerGroup` instances that clear and remount only on scope change, eliminating memory leaks.

Part 2: Executive Actionability & Real-Time Telemetry (Technical + Operations)

Q4: Is this dashboard just read-only analytics, or can an officer execute real statutory actions?

Role: Systems Architect

- **Statutory Bio-Containment Directives:** Officers can open the Outbreak Intelligence modal and execute legally binding containment orders (setting movement bans, market closures, and buffer radiuses).
- **Collector Escalation Channel:** In the Alerts Priority Queue, officers can acknowledge alerts or escalate critical zoonotic outbreaks (Rabies/Anthrax) directly to the District Collector and Disaster Authority.
- **Vaccine Campaign Launchpad:** In Vaccination Intelligence, officers analyze pathogen immunity deficits and launch cold-chain ring-vaccination campaigns targeted strictly at at-risk talukas.
- **Offline SitRep CSV Export:** One-click Situation Report CSV download formats field deployment rosters for Rapid Response Teams during rural power/connectivity outages.

Q5: How does 'Live Telemetry' update without crashing the database under 1,000 concurrent state officers?

Role: Data & API Engineer

- **Stateless REST Architecture:** All dashboard components query stateless `/api/v1` endpoints protected by Spring Security 6 JWT Bearer filters.
- **Redis 7 Micro-Caching:** Outbreak clusters and composite risk decompositions are cached in Redis with a 30-minute TTL; identical officer requests hit RAM cache rather than disk SQL queries.
- **Manual 'Sync Telemetry' Pulse:** Eliminates unthrottled WebSockets battery/server drain; users receive automated status polling with explicit user-triggered refresh capabilities.

Q6: How does administrative scoping work? Can a Dhule District Officer see Pune data?

Role: Public Administration & Policy Analyst

- **Cascading Scope Filter Engine:** Implemented in `scopeFilter.ts` with explicit geographical bounding coordinates, radius math, and district keyword matching across all 7 operational screens.
- **Granular Multi-Tier Scopes:** Supports `Statewide (Maharashtra)`, `Pune District`, `Baramati Taluka`, `Dhule District`, `Jalgaon District`.
- **Role-Based Spatial Containment:** When an officer logs in with district credentials, the scope selector locks to their assigned jurisdiction, preventing unauthorized cross-district data leakage.

Part 3: 5-Year SIH Judge Grilling Scenarios & Proven Defenses (Non-Technical)

Q7: SIH Classic Question: 'How does your dashboard comply with GIGW (Guidelines for Indian Government Websites) and accessibility?'

Role: UI/UX & Accessibility Specialist

• **High-Contrast Semantic Palettes:** Designed to meet WCAG 2.1 Level AA color contrast ratios (Navy `#0E1A2B` on white `#FFFFFF` achieves 14.8:1, well above the 4.5:1 requirement).

• **Color-Blind Safe Symbolism:** Never relies on color alone. Confirmed cases are marked with solid squares (■); suspected cases with hollow diamonds (■); risk tags include text badges.

• **Accessible Tabular Fallback:** Every map visual is backed by an accessible, screen-reader-friendly data table (`OutbreakAccessibleListView.tsx`) for keyboard navigability.

• **Standard Gov Typography:** Utilizes clean, official sans-serif fonts with JetBrains Mono for coordinates and alphanumeric ear-tag IDs.

Q8: SIH Classic Question: 'What prevents a corrupt or panicked officer from shutting down a cattle market on a false AI rumor?'

Role: Healthcare Domain Specialist

• **Two-Man Rule Protocol:** AI-screened animals remain strictly labeled as `SUSPECTED` and cannot legally trigger bio-containment directives on the dashboard.

• **Veterinary Council Authentication:** Containment orders require entering a verified Council Registration Number (VRC) or verified PCR laboratory confirmation from an accredited state LIMS lab.

• **Immutable Access Audit Trail:** All actions (directive deployments, alert acknowledgements, exports) are permanently recorded in the immutable audit ledger with user ID, IP, and timestamp.

Q9: SIH Classic Question: 'Government already has INAPH and Bharat Pashudhan. Why would the Animal Husbandry Ministry deploy this?'

Role: Presenter / Pitcher

• **Retrospective vs. Tactical:** Bharat Pashudhan is a digital census register updated weeks after disease events. Pashu Sathi is a tactical emergency dispatch room.

• **Pre-Configured SOP Library:** Built-in Protocols Reference section houses standard operating procedures (SOPs) for FMD, LSD, Anthrax, and Rabies matching National Disaster Guidelines.

• **Plug-and-Play Integration:** Feeds into existing state infrastructure via open REST APIs without demanding a replacement of state servers or ear-tag hardware.

Executive Summary: The 3-Pillars of the PASHU SATHI Government Command Center

1. Spatial Reality (GIS): Real-time PostGIS clustering compresses outbreak discovery from 14 days of paper reporting down to minutes.
2. Statutory Safety: Mandatory licensed veterinary gates eliminate false alarms and panic shutdowns while adhering strictly to Indian veterinary law.
3. Direct Actionability: Equips district magistrates with instant bio-containment deployment, cold-chain vaccination dispatch, and offline SitRep rosters.